Leveraging Log Probabilities in Language Models to Forecast Future Events

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***Abstract*—In the constantly changing field of data-driven decision making, accurately predicting future events is crucial for strategic planning in various sectors. The emergence of Large Language Models (LLMs) marks a significant advancement in this area, offering advanced tools that utilise extensive text data for prediction. In this industry paper, we introduce a novel method for AI-driven foresight using LLMs. Building on top of previous research, we employ data on current trends and their trajectories for generating forecasts on 15 different topics. Subsequently, we estimate their probabilities via a multi-step approach based on log probabilities. We show we achieve a Brier score of 0.186, meaning a +26% improvement over random chance and a +19% improvement over widely-available AI systems.**

***Index Terms*—artificial intelligence, probabilistic models, large language models, forecasting, foresight, futures studies**

Note

A shorter version of this ongoing work was accepted as a position paper at the 19th IEEE International Conference on Semantic Computing (ICSC 2025) with the title *Anticipating the Future with Large Language Models*.

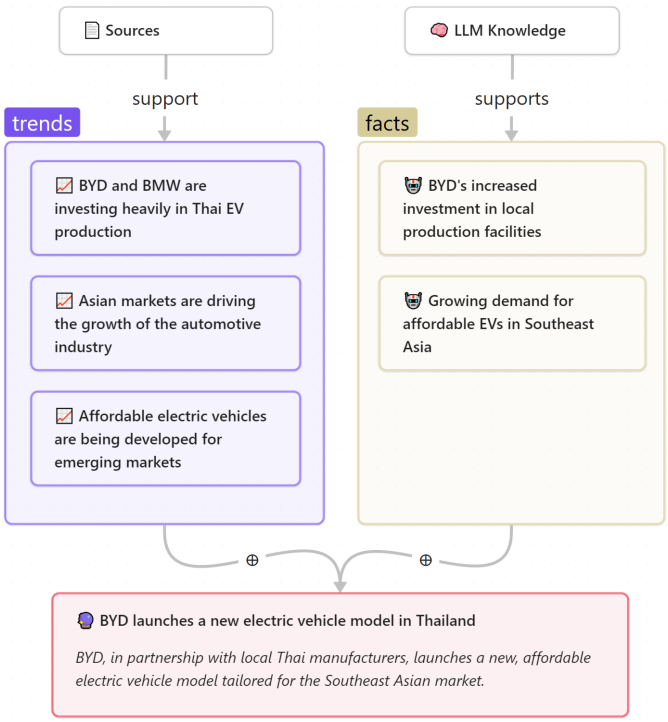

Fig. 1. An example of forecast generation.

I. Introduction

In the constantly changing field of data-driven decision making, accurately predicting future events is crucial for strategic planning in various sectors. The emergence of Large Language Models (LLMs) marks a significant advancement in this area, offering advanced tools that utilise extensive text data for prediction. This paper aims to examine the use of LLMs in forecast generation and probability estimation, emphasising how these models revolutionise the ability to predict future occurrences.

Over recent years, LLMs have demonstrated a remarkable ability to encode and process real-world knowledge within their vast architectures. These models, highlighted by innovations such as Generative Pre-trained Transformers (GPT) [7] and their derivatives, have shown exceptional proficiency in handling various natural language processing (NLP) tasks, from sentiment analysis to predictive modelling. However, their effectiveness in event forecasting remains a contentious issue, as many experts still consider Generative AI "hardly useful for forecasting and decision intelligence" [5]. We aim to challenge this view.

The scope of this paper also extends to the application of LLMs in the context of *Futures Studies* – also known as *Foresight* – a discipline that is highly focused on evaluating the plausibility of different future scenarios. Typically, this task is performed manually by experts through detailed and comprehensive analysis of the world's most transformative issues. Although direct research into the use of LLMs in futures studies is limited, the potential for these models to revolutionise this field is unmistakable. While LLMs have been used recently for event forecasting, these methods presuppose the knowledge of such events and merely estimate their probability. Conversely, ours is the first work towards AI-powered blue-sky forecast generation based on available facts and current trends. We backtest our generated forecasts and show we achieve – in terms of Brier score – a +26%

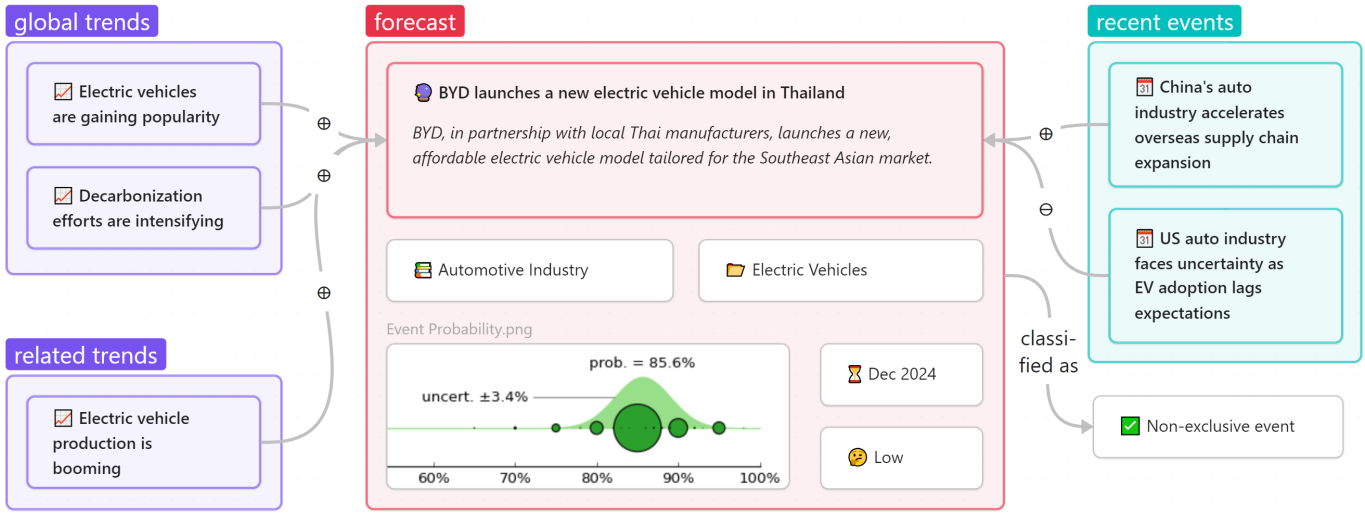

Fig. 2. An example of probability estimation.

improvement over random chance and a +19% improvement over widely-available AI systems.

This paper is organised as follows. Related work is introduced in section II. We describe the approach in section III. We discuss results in section IV. Finally, we conclude.

II. Related Work

The advancement of GPT models has recently transformed natural language processing by introducing effective unsupervised learning methods that generate coherent and contextually-appropriate text [7]. The forecasting capability of these models on diagnosis and triage was tested using healthcare data in [4]. Authors in [1] developed a retrieval-augmented language-model system designed to automatically search for relevant information about a possible future event, perform a forecast, and aggregate predictions. Recently, [8] claimed that LLMs can achieve forecasting accuracy rivalling that of human-crowd forecasting tournaments via an ensemble method of forecast aggregation. *Reasoning and Tools for Forecasting* is a framework of reasoning-and-acting agents that can dynamically retrieve updated information and run numerical simulation with equipped tools [2]. A new AI forecasting system using chain-of-thought was very recently published [6]; unfortunately, its demo is no longer available online at the time of writing this paper.

III. Approach

Our system is composed of three main parts:

1) a **Forecast Generator**, which generates titles, descriptions, and timeframe of potential future events from a topic;
2) a **Probability Estimator**, which outputs a probability, an uncertainty value for a forecast, and related trends;
3) a **Fact Checker**, used to automatically evaluate whether an event has happened within a given timeframe.

In the following, we elaborate on and describe each component in detail.

A. Forecast Generator

The Forecast Generator is an AI-powered foresight tool whose goal is to generate potential future events and their metadata starting from as little as a topic name (e.g., *Automotive*). Building on top of previous research on LLM-driven trend extraction and analysis [10], we collect a broad sample of trends from recent sources. At query time, trends are semantically filtered by relevance to the input topic. This method updates the model's knowledge, which is typically limited to some past cut-off date.

An example of forecast generation is reported in Figure 1. The model generates a forecast based on current trends and facts, which are supported by sources and LLM background knowledge, respectively. The final forecast contains a title and a short description.

For backtesting, we created a dataset of $N = 150$ forecasts from 15 different topics. In order to ensure the absence of leakage of any future data into the dataset, all forecasts were created in February 2024 and consequently fact-checked in October 2024. To compute its probability, each forecast is forwarded to the component introduced in the next subsection.

B. Probability Estimator

Goal of the Probability Estimator is to take a future event as input and output a probability along with a range of values. Unlike previous systems which consider only the LLM's top completion, our system considers all possible guesses for the probability estimation. This way, not only can we estimate a probability $\hat{P}$ as the weighted average of all guesses P_i, but also we can calculate the uncertainty $\hat{U}$ of a given answer as the weighted standard deviation of all guesses:

$$\hat{P} = \frac{\sum_{i=1}^{n} e^{w_i} P_i}{\sum_{i=1}^{n} e^{w_i}} \tag{1}$$

$$\hat{U} = \sqrt{\frac{\sum_{i=1}^{n} e^{w_i} (P_i - \hat{P})^2}{\sum_{i=1}^{n} e^{w_i}}} \tag{2}$$

where w_i is the logarithmic probability (in short, *logprob*) of the token that represents probability P_i. To the best of our knowledge, ours is the first LLM-based forecasting approach that exploits logprobs to compute the final probability value.

Besides their knowledge not being up-to-date, two other limitations of LLMs are their incoherence and inconsistency across runs. *Incoherence* in forecasting happens when two forecasts made for the same event at the same time are assigned different probabilities. To address this issue, we maximise completion determinism by setting temperature and top P to the minimum values allowed by the model. After applying these settings, we noticed that the divergence of probability values across runs remained within 1%. *Inconsistency* can be found when estimating probabilities for events that are mutually exclusive, e.g. elections or sports events. Specifically, we ensure that the sum of the probabilities for all outcomes (e.g., each one of the candidates or competing teams winning) does not exceed 100%. Additional checks may be introduced to avoid ontological inconsistencies, such as having $P(X) < P(Y)$ with X semantically entailed in Y.

The Probability Estimator pipeline expects six steps:

1) reformulate and categorise query, so that the event is spelled correctly in English;
2) search global and local trends relevant to the query;
3) collect sources via semantic search over a news provider;
4) extract the most important events from the headlines;
5) check whether the event is non-exclusive;
6) estimate event probability and uncertainty.

The probability estimation for our running example can be seen in Figure 2. Given the event and timeframe obtained during forecast generation, the model outputs a probability and an uncertainty value, as well as the trends and recent events which positively or negatively affect its probability. Each bubble in the smaller chart represents an estimate of the probability, with its size proportional to the logprob of that estimate.

C. Fact Checker

The Fact Checker component takes a potential event and a timeframe as input. It outputs three possible values: *Happened* (1) if the event has happened, *Inconclusive* (0) if there is no sufficient evidence to confirm the event, or *Did not happen* (-1) if there are reasonable grounds to say that the event cannot possibly have happened. As in the Probability Estimator, relevant sources are collected from our news provider and forwarded to the model to fact-check the input event. We manually evaluated the reliability of the Fact Checker; having achieved a perfect score of $150/150$ on our dataset, it will be employed to track future forecasts automatically.

D. Model calibration

To use the knowledge of forecast uncertainties to our advantage, we learn a transformation function between LLM-returned values and the final probability output. After discarding invalid forecasts, i.e. those events which the Fact Checker found they had already happened at the time of forecasting, we split the dataset equally into training and test set, retaining a test set of $N = 72$ elements. We employ a *Support Vector Regression* model to find the optimal transformation function and ultimately calibrate the model.

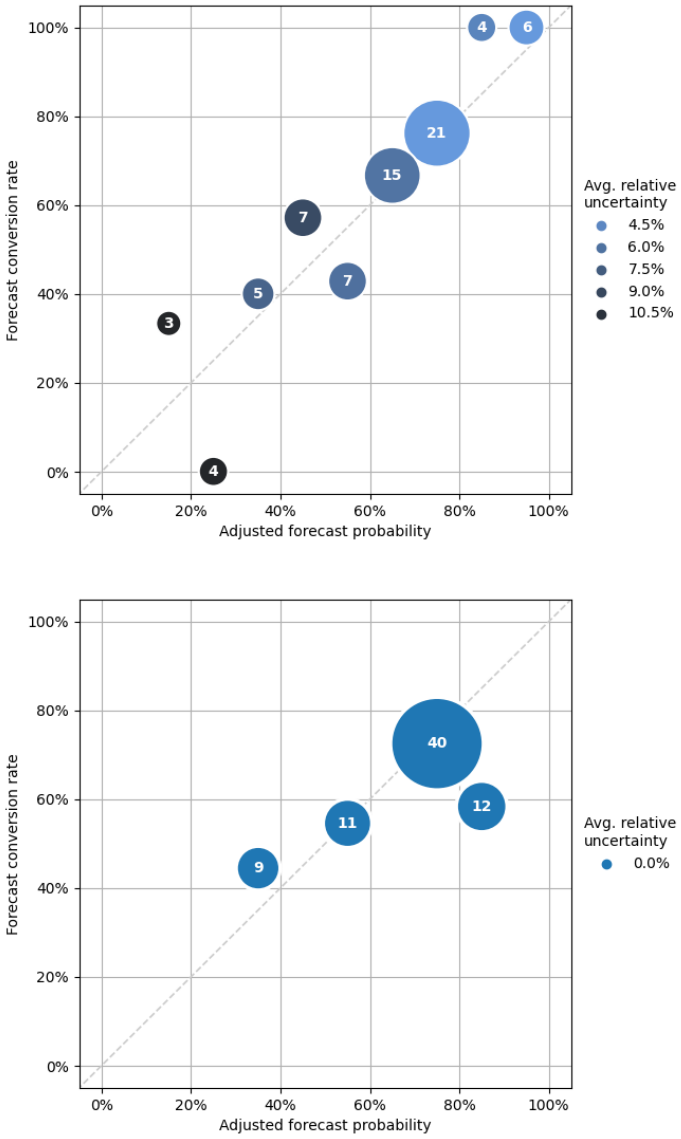

Fig. 3. Calibration plot for our Probability Estimator (above) and vanilla GPT-4o (below).

IV. Evaluation

The trends were extracted and analysed from a collection of ∼45,000 articles over 6 months ending February 2024. As previously mentioned, the time between forecasting and fact-checking is about 240 days. We could not find any system to compare against, therefore we use a widely-available AI as baseline. We employ OpenAI's *GPT-4o* model [3] in both our Probability Estimator and the baseline. As in previous literature [1], [2], [8], we use Brier scores to quantify the performance of our Probability Estimator: